Hamza Ahmed Siddiqui

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PROFESSIONAL SUMMARY

Cloud and Software Engineer with **3+ years** of experience architecting and deploying scalable solutions across **AWS, Azure, and GCP**. Expertise in **microservices development, Infrastructure as Code** (Terraform/CloudFormation), **containerization** (Docker/Kubernetes), and **CI/CD automation** (GitHub Actions/Jenkins). Built production-grade applications including a **cloud-native medication platform** with 8 **microservices on AKS, blockchain-secured medical record system**, and **automated research data pipelines** with **85% improved success rate**. Strong **full-stack foundation** with **Node.js**, **Python, React**, and multiple database technologies. Passionate about solving complex problems with elegant, scalable, and secure architectures.

SKILLS & TOOLS

Cloud Platforms: AWS (EC2, S3, Lambda, Athena, Glue, CloudWatch, ECS, Azure (AKS, App Services) , Google Cloud Platform, Firebase (Auth, Firestore, Functions, Hosting)
Infrastructure & DevOps: Kubernetes, Docker, Terraform, AWS CloudFormation, GitHub Actions, Jenkins, Ansible, Helm, Prometheus, Grafana
CI/CD & Version Control: Git, GitHub, GitLab, Mercurial, CI/CD Pipeline Design, Automated Testing, Build Automation, Release Management
Programming Languages: JavaScript (ES6+), TypeScript, Python, Node.js, Java, C++, HTML5, CSS3, Solidity (Basic)
Frameworks & Libraries: React.js, Next.js, Express.js, Spring Boot, Django, D3.js, OpenLayers, Puppeteer, Cheerio, jQuery
Databases: PostgreSQL, MySQL, MongoDB, Firebase Firestore, DynamoDB, Redis, Elasticsearch, SQLite
APIs & Architecture: RESTful APIs, GraphQL, WebSockets, Microservices, Event-Driven Architecture, JWT, OAuth 2.0, Web3.js
Security & Compliance: AES-256 Encryption, HIPAA/GDPR Compliance, Role-Based Access Control (RBAC), JWT Authentication, Firebase Auth with 2FA, Rate Limiting, Input Validation
Data Engineering & Analytics: AWS Athena, AWS Glue, Amazon QuickSight, ETL Pipelines, Web Scraping, Data Visualization, CSV/Excel Processing
Monitoring & Observability: Prometheus, Grafana, AWS CloudWatch, Firebase Performance Monitoring, Logging, Error Tracking, Uptime Monitoring
Development Tools: VS Code, Postman, Docker Desktop, Kubernetes CLI (kubectl), Terraform CLI, NPM, Yarn, Webpack, Babel
Methodologies: Agile (Scrum), DevOps, Infrastructure as Code (IaC), Test-Driven Development (TDD), CI/CD, Git Flow

PROFESSIONAL EXPERIENCE

Ecomtent - Cloud Developer | *Toronto, ON* — Jan 2025 - March 2025

- Designed and deployed **cloud-native BI pipelines** using **AWS Athena, Glue, S3, and QuickSight** enabling real-time analytics that improved business decision velocity by **25%** through automated data processing and optimized SQL strategies
- Built **serverless data pipelines** with automated infrastructure using **AWS CloudFormation** while optimizing partitioning strategies that reduced dashboard load time by **30%** for end users across the organization
- Implemented **data quality checks and pipeline audits** using **AWS Glue and Lambda** improving reporting accuracy by **20%** across all BI dashboards while architecting event-driven workflows for real-time data ingestion
- Established **comprehensive monitoring and alerting** with **AWS CloudWatch** ensuring **99.9% pipeline reliability** while documenting architecture and training team members on best practices for cloud data engineering

Chemical 101 - Software Engineer | *Toronto, ON* — Oct 2023 - Dec 2023

- Built **cloud-hosted Node.js ingestion service** for automated data scraping using **Puppeteer and Cheerio** with proxy rotation and retry logic improving system reliability and success rate by **35%**
- Created **CI/CD pipelines using GitHub Actions** for automated testing and deployment while implementing asynchronous programming and modular architecture for optimal performance across distributed environments
- Improved data ingestion efficiency through **optimized parallel processing and error handling** reducing manual workload by **40%** and enabling 24/7 unattended data collection from multiple sources
- Designed **RESTful APIs** for data access with comprehensive logging and monitoring ensuring high availability and quick issue resolution during production operations

Dematic Software - Software Engineer | *Stockport, England* **Sep 2021 – May 2022**

- Developed and maintained **internal cloud-backed microservices** supporting **500+ users** with **Spring Boot and Node.js** while managing PostgreSQL and MySQL infrastructure using **Terraform** for Infrastructure as Code
- Built secure **REST APIs** with optimized data flows implementing caching strategies and query optimization that reduced support resolution time by **35%** for internal and external stakeholder requests
- Migrated **source control from Mercurial to Git with** comprehensive branch strategies and automated workflows improving deployment reliability by **25%** and enabling better team collaboration
- Worked in **Agile DevOps teams** with **CI/CD pipelines using Jenkins** and automated testing achieving faster release cycles and higher code quality standards across all services

PROJECTS

MediTrack - Cloud-Native Medication Management Platform | *Capstone Project*
Technologies: Node.js, Python, Firebase, Kubernetes, Docker, Terraform, React, GitHub Actions

- Architected and deployed **8 microservices** on **Azure Kubernetes Service** handling **500+ concurrent users** with real-time medication reminders and caregiver collaboration using **Firebase Firestore** and **WebSocket** connections
- Implemented **Infrastructure as Code with Terraform** reducing deployment time by **60%** and built **CI/CD pipelines with GitHub Actions** achieving **99.9% deployment success rate** through automated testing and containerized deployments.
- Developed **Python-based medicine scraper** aggregating data from **NHS/CDC databases** with **Redis caching** reducing API response time by **45%** while implementing **HIPAA-compliant security** with **Firebase Auth 2FA and RBAC**

MediVault - Secured Medical Records Platform | *Personal Project*
Technologies: React, Node.js, MongoDB, AES-256, JWT, Tailwind CSS

- Engineered **HIPAA/GDPR-compliant** medical record system with **AES-256 encryption** and **JWT authentication** ensuring secure role-based access control for patients and healthcare providers
- Designed hybrid architecture storing encrypted records in **MongoDB** with comprehensive audit logging while implementing patient-centric access control allowing individuals to grant or revoke provider permissions
- Built **React.js frontend** with **Tailwind CSS** featuring intuitive dashboards and developed **secure Node.js backend** with input validation, rate limiting, and encrypted data storage meeting healthcare compliance requirements

Google Scholar Scraper - Automated Research Data Pipeline | *Personal Project*
Technologies: Node.js, Puppeteer, Python, Express, Docker, GitHub Actions

- Engineered **headless browser automation** with **Puppeteer** extracting **500+ academic records** using proxy rotation and **CAPTCHA detection** with automatic retries increasing data collection success rate by **85%**
- Built configurable data pipelines exporting to **CSV and Excel** with keyword filters and geographic targeting via **Google Geocoding API**, containerized with **Docker** for consistent execution across environments
- Implemented comprehensive error handling with retry logic and rate-limiting safeguards while creating **CI/CD pipelines with GitHub Actions** enabling **24/7 unattended** data collection for research analysis

Siddiqui Family Tree - Interactive Genealogy Visualization | *Personal Project*
Technologies: JavaScript, D3.js, OpenLayers, HTML5, CSS3, JSON

- Designed interactive family tree using **D3.js force-directed graphs** displaying **20+ family members** across multiple generations with zoom/pan, hover highlighting, and click-to-center navigation enhancing user engagement by **40%**
- Integrated **OpenStreetMap via OpenLayers** to geolocate family heritage sites across different countries adding spatial storytelling dimension while structuring data in **JSON format** with comprehensive relationship mapping
- Created **responsive SVG graphics** maintaining clarity across devices and deployed as **static web application with zero server costs** demonstrating rich client-side experiences without backend dependencies

EDUCATION & CERTIFICATIONS

Graduate Certificate in Cloud Architecture & Administration | Seneca College, Toronto, ON 2025
Graduate Certificate in Computer Application Development | Conestoga College, Kitchener, ON 2024
B.Eng. in Robotic Engineering| University of Essex, Colchester, UK 2021

AWS Certified Cloud Practitioner | CCNA Routing & Switching | GraphQL API | Apollo Server | AWS Academy Cloud Foundations | C++ | Architecting Serverless Applications | ROS | RPA | AI